

TROY 2 W, AL, USA

WMO: 750010

Lat: 31.7901N

Lon: 86.0004W

Elev: 472

StdP: 14.45

Time Zone: -6.00 (NAC)

Period: 08-19

WBAN: 23801

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 23.5	(c) 27.3	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 16.5	(k) 65.0	(l) 14.0	(m) 61.0	(n) 3.7	(o) 10	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 18.8	(c) 93.4	(d) N/A	(e) 91.7	(f) N/A	(g) 90.1	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 4.8	(p) 300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 12.6	(b) 10.8	(c) 9.3	(e) 17.6	(f) 95.7	(g) 5.0	(h) 2.3	(i) 14.0	(j) 97.4	(k) 11.1	(l) 98.7	(m) 8.3	(n) 100.0	(o) 4.7	(p) 101.7		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	64.8	47.0	52.0	57.6	64.4	71.5	78.0	79.6	79.1	75.7	65.7	54.9	51.3
	DBStd	13.56	10.72	9.74	8.80	6.50	5.54	3.08	2.78	2.93	4.71	7.78	8.36	9.77
	HDD50	465	188	87	35	1	0	0	0	0	0	3	42	109
	HDD65	2135	561	372	255	90	18	0	0	0	2	88	317	432
	CDD50	5873	95	144	271	434	667	841	916	903	772	488	191	151
	CDD65	2068	3	10	26	72	220	391	451	438	324	109	15	9
	CDH74	17518	4	34	180	620	1919	3568	3982	3681	2600	848	67	15
	CDH80	6929	0	1	16	101	646	1584	1751	1570	1049	205	6	0
Wind	WSAvg	4.1	5.0	5.4	5.4	4.9	3.8	3.6	3.2	3.3	3.1	3.9	3.6	4.6
Precipitation	PrecAvg	55.7	4.9	4.8	6.3	4.4	3.6	5.0	6.0	4.5	3.7	2.9	4.2	4.6
	PrecMax	90.9	10.7	11.9	15.4	9.2	10.4	12.6	15.6	11.3	13.3	12.2	10.6	16.3
	PrecMin	40.6	0.9	0.9	0.4	0.7	0.1	1.0	1.7	0.6	0.6	0.0	1.2	1.2
	PrecStd	8.9	2.3	2.6	3.3	2.4	2.1	2.8	3.1	2.4	2.5	2.7	1.9	2.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	74.5	78.1	82.5	85.6	91.5	95.5	95.0	95.5	94.2	88.6	80.2	76.3
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	70.6	74.2	78.8	83.1	88.7	93.1	92.4	92.5	91.3	85.0	75.5	72.7
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	67.6	70.4	75.6	80.6	86.3	90.9	90.7	90.4	89.0	82.1	72.6	69.7
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	63.9	67.6	72.1	78.0	83.9	88.7	88.9	88.3	86.5	78.9	69.8	67.0
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Mean Daily Temperature Range	5% DB	MDBR	20.8	21.3	22.1	22.7	21.9	20.0	18.8	18.3	19.6	22.3	23.2	19.0
		MCDBR	20.4	21.2	23.4	24.5	24.8	23.8	21.0	21.4	23.1	23.2	23.1	18.7
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Clear-Sky Solar Irradiance	taub	0.321	0.329	0.354	0.381	0.429	0.470	0.520	0.507	0.426	0.368	0.344	0.325	
	taud	2.531	2.511	2.442	2.366	2.267	2.209	2.077	2.113	2.320	2.458	2.470	2.533	
	Ebn at Noon	287	296	294	289	274	261	248	250	267	278	275	280	
	Edh at Noon	27	30	35	39	43	46	52	50	39	31	29	26	
All-Sky Solar Radiation	RadAvg	918	1099	1476	1824	2000	1941	1893	1755	1571	1351	1036	792	
	RadStd	59	137	127	145	140	152	121	108	124	160	92	86	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (44 neighbors)		+0.86	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+297	+193

Nomenclature: See separate page